

Principal Component Analysis I

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Recap

- Last time we covered **Random Forests**:
 - Procedure.
 - Hyperparameters and tuning.
 - Variable importance.

Introduction

- It can be very challenging to effectively work with many correlated variables.
- May be able to use variable selection methods to reduce the number of variables.
- May also be able to use dimension reductions maintain the important structures in the data.
 - **Principal Component Analysis (PCA)**
 - Canonical Correlation Analysis (CCA)
 - Singular Value Decomposition (SVD)
 - Factor Analysis (FA)

Dimension Reduction

- **Dimension reduction (ordination)** techniques are especially useful when there are many correlated variables.
- Can be seen as a way to *approximate high-dimensional data* with a few lower dimensional matrices.
- These lower dimensional matrices may then be used for further analysis.

Illustrative Example 1

- ➊ Run the code to simulate a matrix with ten columns of normally distributed data.
 - ➊ Use the `heatmap()` function to examine your sample.
 - ➋ What do you notice?

- ➋ Run the second code chunk to generate a matrix that adds a pattern to the data from part 1.
 - ➊ Use the `heatmap()` function to examine your sample.
 - ➋ What do you notice?

Patterns in Data

- In our data, there may be patterns that exist across rows and columns.
- These patterns may not be obvious at first glance or in the case of many dimensions.
- The patterns may actually be layered on top of each other.

Illustrative Example 2

- 1 Run the code to further examine the matrix from Illustrative Example 1.
- 2 What do you notice?

Principal Component Analysis

Principal Component Analysis

- **Principal component analysis (PCA)** is a method for finding low-dimensional representations of a dataset.
 - Retain as much of the original variation as possible.
- Each of our n observations exist in p -dimensional space, but not all of the dimensions are meaningful.
 - Interested in a smaller number of *meaningful* dimensions.
- Each of the *new* dimensions is a linear combination of the original p features.
 - Eigenanalysis-based approach.

Eigenanalysis Review

- **Eigenanalysis** is a method used to identify a set of linear equations that summarize a square matrix.
- The approach yields a set of **eigenvalues** (λ) which have an associated **eigenvector** ($\mathbf{X}$):

$$\mathbf{AX} = \lambda\mathbf{X} \quad (1)$$

- The multiplication of a square matrix $\mathbf{A}$ and a vector $\mathbf{X}$ will yield the same values as the multiplication of a scalar value λ and the vector $\mathbf{X}$.
- This allows us to manipulate our individual data points in a coordinate-space by multiplying them by an eigenvector $\mathbf{X}$.
 - *Very important for PCA.*

Eigenanalysis

- Each dimension of a square matrix is represented by an eigenvector and associated eigenvalue.
- The dimensions are decreasing in order of importance.
 - Allowing us to focus on the first few dimensions and still see broad patterns in our data.
- If all eigenvectors are used, the patterns in the data are perfectly preserved.
- A process of identifying and computing a new coordinate system for the data.
 - **Axes of the new coordinate system are uncorrelated with one another.**

PCA Explanation

- Each eigenvalue measures the amount of variance explained by that principal component.
- The first eigenvalue accounts for as much variation as possible, the second as much of the remaining variation as possible, and so on.
- Each eigenvector is the set of weights assigned to the original variables to produce that principal component.
- Each principal component is uncorrelated with the others.
- *PCA identifies a series of new 'synthetic' variables (principal components; PCs) that are composites or blends of the original variables.*

Illustrative Example 3

- 1 Run the provided code to generate a data cloud with principal components.
- 2 What do you notice?

PCA Procedure

- ➊ Normalize your data (optional).
- ➋ Calculate the covariance matrix **or** correlation matrix.
 - Covariance matrix with normalized data.
 - Correlation with non-normalized data.
- ➌ Conduct Eigenanalysis.
- ➍ Interpret the Eigenvalues.
- ➎ Interpret the Eigenvectors.
- ➏ Interpret the Scores of Sample Units

Interpreting Loadings

- The threshold of importance varies across literature.
- A variable is most important for the principal component where its loading has the largest absolute value.
- Larger magnitude = more weight
- Similar magnitude and same direction = similar weight (both variables increase at same time)
- Similar magnitude but opposite direction = contrasting weight (one variable increases as the other decreases)
- *The process contains some subjectivity.*

Interpreting Scores

- The **scores** represent each object's position in the *newly defined* coordinate space.
- The scores are calculated by multiplying the original scaled data matrix by the loadings (eigenvectors).
- The PC scores are orthogonal to (uncorrelated with) each other.

Example 1

- Import the *Basketball.csv* dataset to do the following:
 - ① Normalize your data (optional).
 - ② Use the provided code to filter out categorical variables and calculate the correlation matrix.
 - **In general, you can use one-hot encoding for categorical variables.**
 - ③ Conduct Eigenanalysis.
 - ④ Interpret the Eigenvalues.
 - ⑤ Interpret the Eigenvectors.
 - The elements of an eigenvector are known as the **loadings**.
 - ⑥ Interpret the Scores of Sample Units.

Next Lecture

- We will discuss **PCA II**.
- Number of principal components.
- Easier implementation in R.
- Some applications

Exercise 1

- Import the wine dataset from the *HDclassif* package.
- Normalize your data (optional).
- Filter out categorical variables and calculate the correlation matrix.
- Conduct Eigenanalysis.
- Interpret the Eigenvalues.
- Interpret the Eigenvectors.
- Interpret the Scores of Sample Units.
- What do you notice?

References & Resources

- ① Bakker, J. D. (2026) *Applied Multivariate Statistics in R*
 - ② Peng R. D. (2020) *Exploratory Data Analysis with R*
 - ③ Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction*. Chapman and Hall/CRC
 - ④ Bradley Boehmke B., and Greenwell B. (2020) *Hands-On Machine Learning with R*. Taylor & Francis
- `eigen()`